

Python Assignment Submission - Nexgensis Tech

Please use this form to submit your solution and provide context on your implementation. As mentioned in the assignment notes, if you encountered any ambiguous scenarios, we are interested in **how you resolved them**. Please explicitly state your logic and assumptions in the relevant sections below.

Important:

1. Ensure your GitHub link is **public** or accessible.
2. Double-check that your report.json is included in your repository.
3. Your submission will be evaluated on code quality, logic, and your ability to document your engineering decisions

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GitHub Repository Link *

[https://github.com/palash27114/Mystery-Deli](https://github.com/palash27114/Mystery-Delivery)

Describe the most significant technical challenge you encountered during this assignment. How did you resolve it? *

The main technical challenge was handling dynamic fleet locations and preventing calculation errors in edge cases. Since an agent's location changes after every delivery, the solution maintains the current position of each agent and updates it immediately after every delivery. It also prevents division-by-zero errors by assigning an efficiency of 0.0 to agents who deliver no packages and excluding them from best-agent selection. In addition, a strict `validate_delivery_data` schema validates coordinate values and ensures that all referenced warehouses exist before the simulation begins, preventing invalid input from causing failures.

Please provide a brief summary of your approach *

The solution uses a modular Python architecture in `src/fastbox/`, separating validation, distance calculation, simulation, analytics, and reporting into different modules. The dispatch engine processes packages sequentially in FIFO order, calculates the Euclidean distance from each agent's current location to the pickup warehouse, assigns the package to the nearest available agent, and then moves that agent to the package destination. Fleet analytics calculates each agent's efficiency as total distance travelled divided by packages delivered, with a lower value representing better efficiency, and identifies the best-performing eligible agent.



Since the assignment allowed for interpretation, please list the specific logic assumptions you made (e.g., how you handled tie-breaking or routing order).

The simulation follows FIFO order, meaning packages are dispatched in exactly the order provided in the input dataset. Each delivery is treated as a separate trip in which the agent travels from its current location to the package's warehouse and then to the package destination, remaining there until the next assignment. If multiple agents are equally close to a warehouse, the tie is resolved deterministically using alphabetical or dictionary-key order. Efficiency is interpreted as average travel distance per delivered package, where a lower value indicates greater efficiency. Only agents who deliver at least one package are eligible to be selected as the best agent, while agents with zero deliveries are excluded. All agents are assumed to start the simulation simultaneously at their specified initial coordinates.

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